A

PROJECT REPORT

ON

Autonomous Cybersecurity: AutoML-Based Intrusion Detection System

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF TECHNOLOGY

in

ARTIFICIAL INTELLIGENCE AND DATA ENGINEERING

Submitted By:

Shravan Deshpande (2023UAI1894)

Submitted To:

Dr. Ramesh Babu Battula

Associate Professor

Abstract

This project presents a semi-autonomous Intrusion Detection System (IDS) using AutoML techniques for cybersecurity applications. The framework uses tree-based machine learning models such as Decision Tree, Random Forest, Extra Trees, XGBoost, LightGBM, and CatBoost. Automated feature selection is performed using averaged feature importance scores from all models, followed by TVAE-based synthetic data generation for balancing minority attack classes. Ensemble learning and confidence-based stacking are used to improve robustness and classification accuracy.

1. Introduction

Modern 5G/6G networks and IoT systems generate massive network traffic, making cybersecurity increasingly important. Traditional IDS systems require heavy manual feature engineering and hyperparameter tuning. This project aims to automate several stages of intrusion detection using AutoML techniques.

2. Objectives

- Develop an intelligent intrusion detection framework.
- Reduce manual feature engineering and tuning.
- Improve attack detection performance using ensemble learning.
- Apply AutoML concepts to cybersecurity.
- Handle imbalanced datasets using TVAE balancing.

3. Datasets Used

The project uses CICIDS2017 and 5G-NIDD datasets containing both benign and malicious traffic records. Each row represents a network flow and each column represents a network feature.

3.1 Important Features

- Flow Duration
- Flow Bytes/s
- Flow Packets/s
- Packet Length Mean
- SYN Flag Count
- ACK Flag Count
- Inter Arrival Time (IAT)
- Active Mean and Idle Mean

4. Overall Workflow

1. Dataset Loading
2. Data Preprocessing
3. Training 6 Base Tree Models
4. Model Evaluation
5. Automated Feature Selection

6. TVAE-Based Data Balancing
7. Retraining Optimized Models
8. Ensemble Learning and Stacking
9. Final IDS Prediction

5. Data Preprocessing

The preprocessing stage handles missing values, removes infinite values, encodes attack labels numerically, and splits the dataset into training and testing sets.

6. Machine Learning Models Used

Decision Tree: Baseline rule-based classifier.

Random Forest: Ensemble of decision trees.

Extra Trees: Randomized ensemble tree model.

XGBoost: Gradient boosting with regularization.

LightGBM: Fast leaf-wise boosting model.

CatBoost: Ordered boosting with improved generalization.

7. Understanding Boosting Algorithms

Boosting algorithms train trees sequentially. Each new tree learns the errors made by previous trees. This iterative learning significantly improves prediction performance.

7.1 XGBoost

XGBoost uses gradient descent optimization and regularization to reduce overfitting while learning complex nonlinear attack patterns.

7.2 LightGBM

LightGBM uses leaf-wise tree growth and histogram-based splitting, making it highly efficient for large cybersecurity datasets.

7.3 CatBoost

CatBoost uses ordered boosting and efficiently handles categorical relationships, reducing prediction bias.

8. Automated Feature Selection

Feature importance values are extracted from all six tree models. The importance scores are averaged and sorted. Features are selected until cumulative importance reaches 90%, while low-importance features are discarded.

9. TVAE-Based Data Balancing

TVAE generates synthetic minority attack samples by learning the statistical distribution of attack classes. This improves rare attack detection performance.

10. Ensemble Learning

The framework combines predictions from top-performing models using stacking and confidence-based ensemble learning to improve robustness and accuracy.

11. Project Contributions

- Semi-autonomous IDS framework
- Automated feature selection
- TVAE-based balancing
- Confidence-based ensemble stacking
- Designed for 5G/6G cybersecurity

12. Conclusion

This project demonstrates a semi-autonomous cybersecurity framework integrating AutoML concepts, feature selection, boosting algorithms, TVAE balancing, and ensemble learning for robust intrusion detection.